

Department of Mechanical Engineering

Course Outcomes of all courses of B Tech 5th semester MECHANICAL ENGG.

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C301 Internal Combustion Engine C037511(037)	C 301.1	Describe the construction and working principle of various internal combustion engines. Explain the concepts of fuel air cycle and actual cycle and apply it to analyze related practical problems. (LEVEL 2,3)
	C 301.2	Explain the theory of combustion of S.I. engine and C.I. engine, describe I.C.Engine fuels and solve problem related to flue gas analysis. (LEVEL 2,3)
	C 301.3	Discuss properties of air-petrol mixtures and describe fuel supply system of S.I. and C.I. Engine. (LEVEL 4)
	C 301.4	Describe ignition system, cooling system, lubrication system and Engine emissions and its control. (LEVEL 3)
	C 301.5	Describe various performance parameter of I.C. Engine, its method of testing and analyze related practical problems. (LEVEL 3)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C302- Solid Mechanics C037512(037)	C302.1	Analyze problems related to deformable body under load using energy methods. (LEVEL 3)
	C302.2	Analyze fixed beams and continuous beams under load. (LEVEL 3)
	C302.3	Analyze thin and thick pressure vessels. (LEVEL 3)
	C302.4	Analyze column and find shear center. (LEVEL 3)
	C302.5	Solve plane stress and plain strain problems. (LEVEL 3,4)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C303- Fluid Machines C037513(037)	C303.1	Explain the concepts of ‘boundary layer theory’ and ‘ lift and drag theory’ and apply to solve related practical problems (LEVEL 2,3)
	C303.2	Explain the principle of impulse-momentum and impulse turbines and apply it to analyze related problems. (LEVEL 2,3)
	C303.3	Explain the construction and principle of operation of reaction turbine and apply it to analyze related problems. (LEVEL 2,3)
	C303.4	Explain the construction and principle of operation of centrifugal pump and apply it to analyze related problems. (LEVEL 2,3)
	C303.5	Explain the construction and principles of operation of reciprocating pump and apply it to analyze related problems. (LEVEL 2,3)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C304-Dynamics of Machine C037514(037)	C304.1	Explain principles of operation of mechanical governors and analyze its performance parameters. (LEVEL 2,3)
	C304.2	Apply the theory of balancing to rotating and reciprocating masses. (LEVEL 3)
	C304.3	Analyze gyro-effect on moving bodies. (LEVEL 4)
	C304.4	Explain principles of vibrations of different systems and analyze related practical problems. (LEVEL 2,3)
	C 304.5	Perform inertia force analysis of machine elements. Draw turning moment diagram of reciprocating engine and analyze performance parameter of flywheel. (LEVEL 3)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C305- Operation Research C037531(037)	C305.1	Formulate and solve real-world problems as linear programs for better decision-making. (LEVEL 3,4)
	C305.2	Solve specialized linear programming models like the transportation and assignment Models. (LEVEL 3,4)
	C305.3	Model a dynamic system as a queuing model and compute important performance measures. (LEVEL 3)
	C305.4	Use CPM and PERT techniques, to plan, schedule and control project activities. (LEVEL 3)
	C305.5	Propose the best strategy using decision making methods under game theory & apply concepts of Simulation to optimize practical problems. (LEVEL 6,3)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C306- Internal Combustion Engine Lab C037521(037)	C306.1	Describe the basic engine nomenclature and working principle of four stroke and two stroke Petrol and Diesel engine. .(LEVEL 2,3)
	C306.2	Describe the fuel supply system of a Petrol and Diesel engine. .(LEVEL 2,3)
	C306.3	Describe Ignition, Lubrication and cooling system of an internal combustion engine. .(LEVEL 3)
	C306.4	Analyze the performance parameters of diesel engine. .(LEVEL 3)
	C306.5	Analyze the performance parameters of petrol engine. .(LEVEL 3)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C307- Dynamics of Machine Lab C037522(037)	C307.1	Analyze the vibration parameters of various systems. .(LEVEL 3)
	C307.2	Analyze gyroscopic parameters. .(LEVEL 3)
	C307.3	Analyze various types of governors. .(LEVEL 3)
	C307.4	Find the critical speed of different diameters of shafts. .(LEVEL 1,2)
	C307.5	Analyze the effects of unbalance in machine and methods to reduce/eliminate these effects. .(LEVEL 3)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C308- Fluid Machines Lab C037523(037)	C308.1	Analyze the performance parameters of Pelton Turbine. .(LEVEL 3)
	C308.2	Analyze the performance parameters of Francis and Kaplan Turbine. (LEVEL 3)
	C308.3	Analyze the performance parameters of Centrifugal Pump and Reciprocating Pump. .(LEVEL 3)
	C308.4	Determine Lift and drag force over an air foil. .(LEVEL 4)
	C308.5	Explain the construction and working of various fluidic devices. .(LEVEL 2)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C309- Project-I based on Summer Internship/ Industrial Training C037524(037)	C309.1	Technical Skills Development Students will be tasked with applying the technical skills they've acquired during their internship or training to solve real-world problems. This could involve implementing software solutions, conducting experiments, or troubleshooting technical issues encountered in the industry. (Level: 3)
	C309.2	Problem-Solving Abilities Students will analyze complex problems relevant to their field of study or industry. They will identify key issues, examine different approaches to problem-solving, and propose innovative solutions. This could involve case studies, simulations, or research-based projects. (Level: 4)
	C309.3	Communication Skills Enhancement Students will evaluate and improve their communication skills through project presentations, reports, or documentation. They will articulate their ideas, present findings, and defend their solutions effectively. Peer reviews and feedback sessions can be incorporated to enhance communication abilities. (Level: 5)
	C309.4	Professional Development Students will create a professional portfolio showcasing their achievements, experiences, and skills gained during the internship or training. They will reflect on their learning journey, set career goals, and develop a plan for continuous professional development. This could involve creating resumes, LinkedIn profiles, or personal branding materials. (Level: 6)
	C309.5	Teamwork and Collaboration Students will apply their teamwork and collaboration skills by working on group projects or collaborative assignments. They will demonstrate effective teamwork, leadership, and interpersonal skills while completing tasks and achieving project objectives. Group presentations or joint reports can be used to assess teamwork outcomes. (Level: 3)